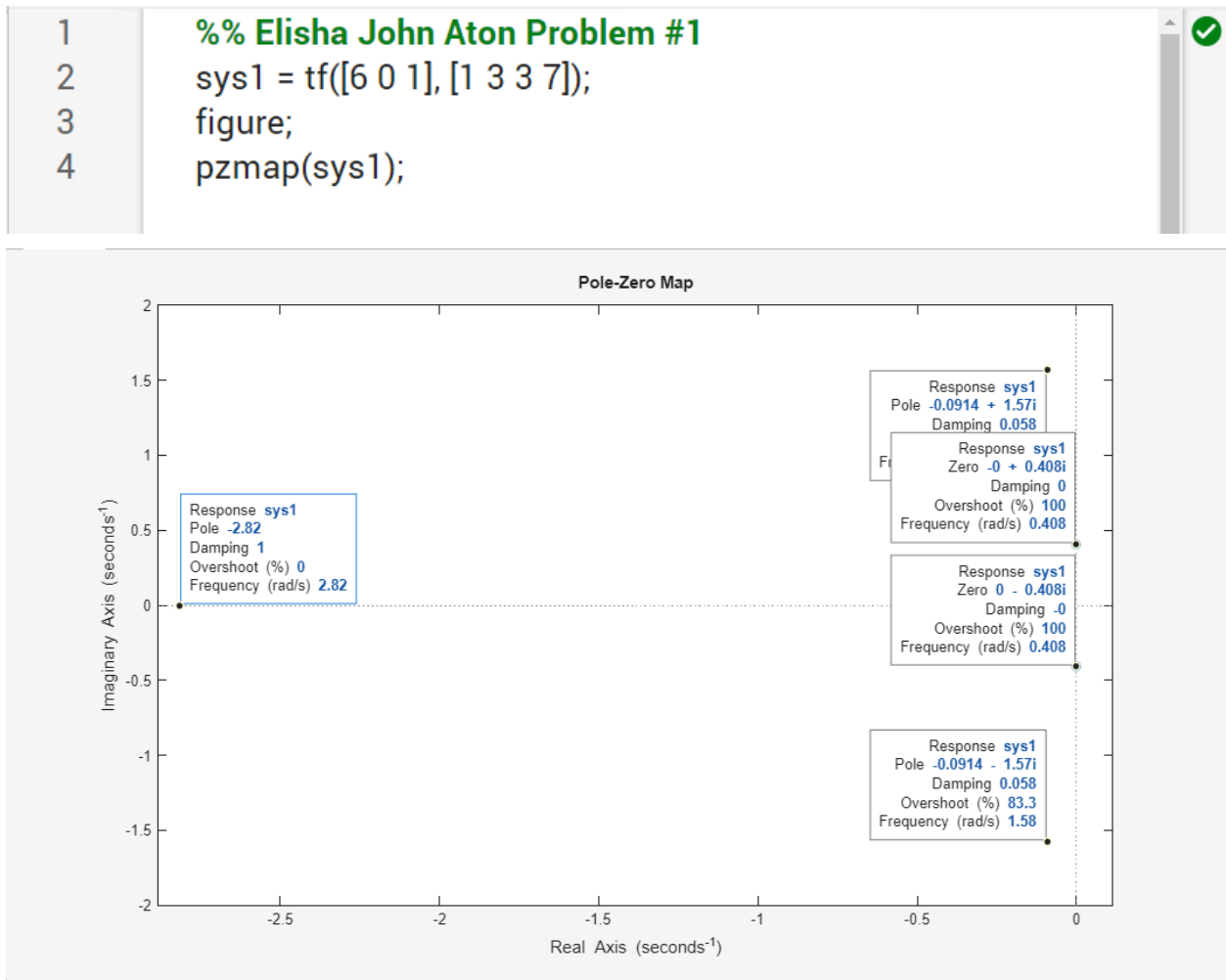
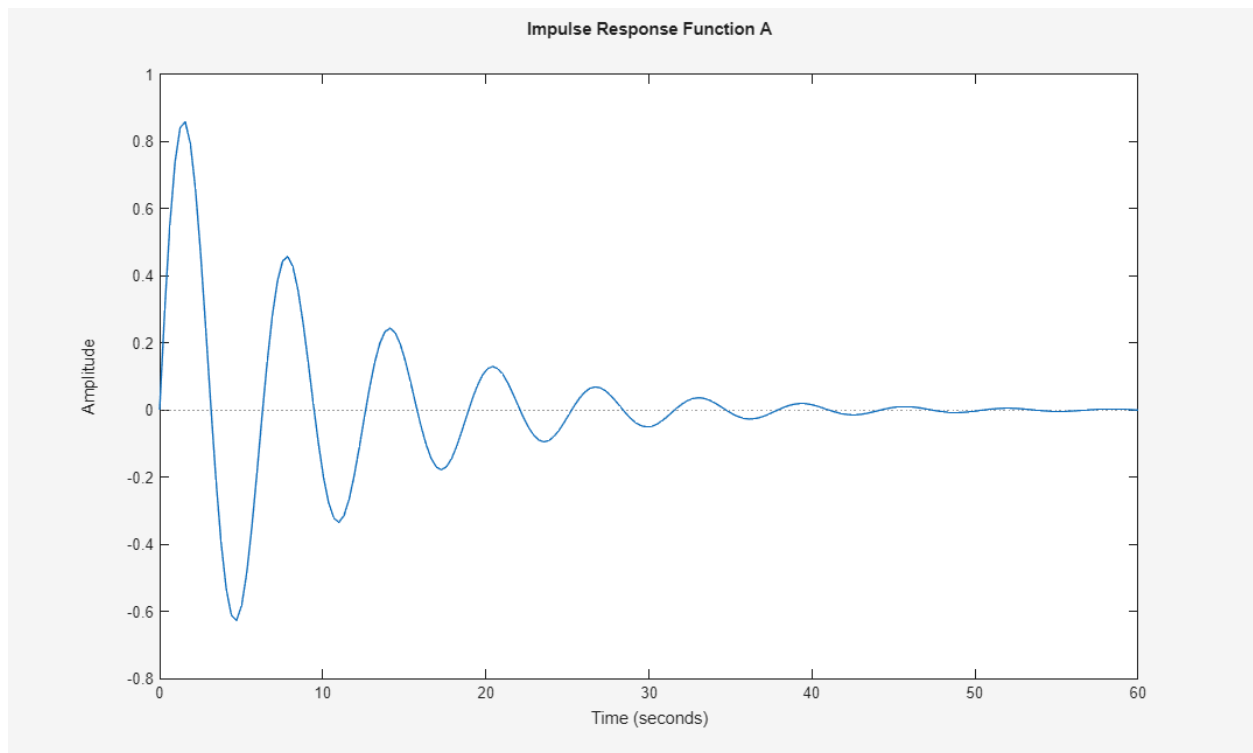


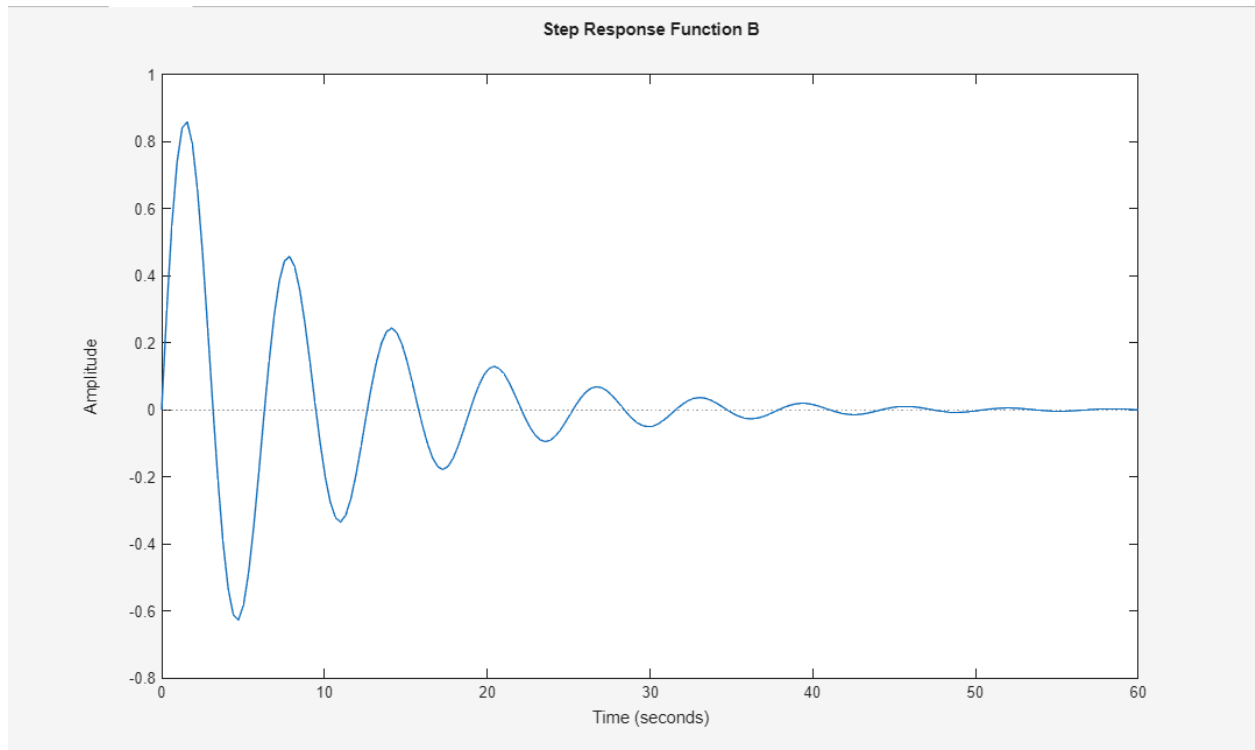
1.



2.

```
1 %% Elisha John D. Aton Prob #2
2 eqn1 = tf(1, [1 0.2 1]);
3 eqn2 = tf([1 0], [1 0.2 1]);
4
5 figure;
6 impulse(eqn1);
7 title('Impulse Response Equation 1');
8 figure;
9 step(eqn2);
10 title('Step Response Equation 2');
```





when calculating the unit impulse response of a transfer function, multiply by 1.  
for unit step response, multiply by  $1/s$

obtaining unit impulse response for transfer function 1

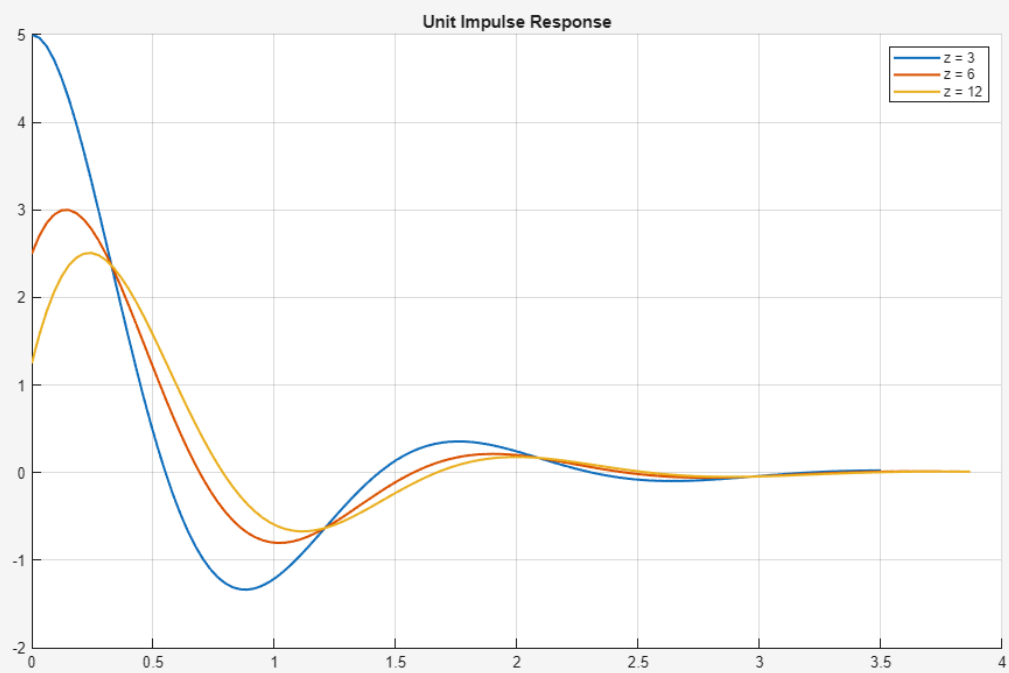
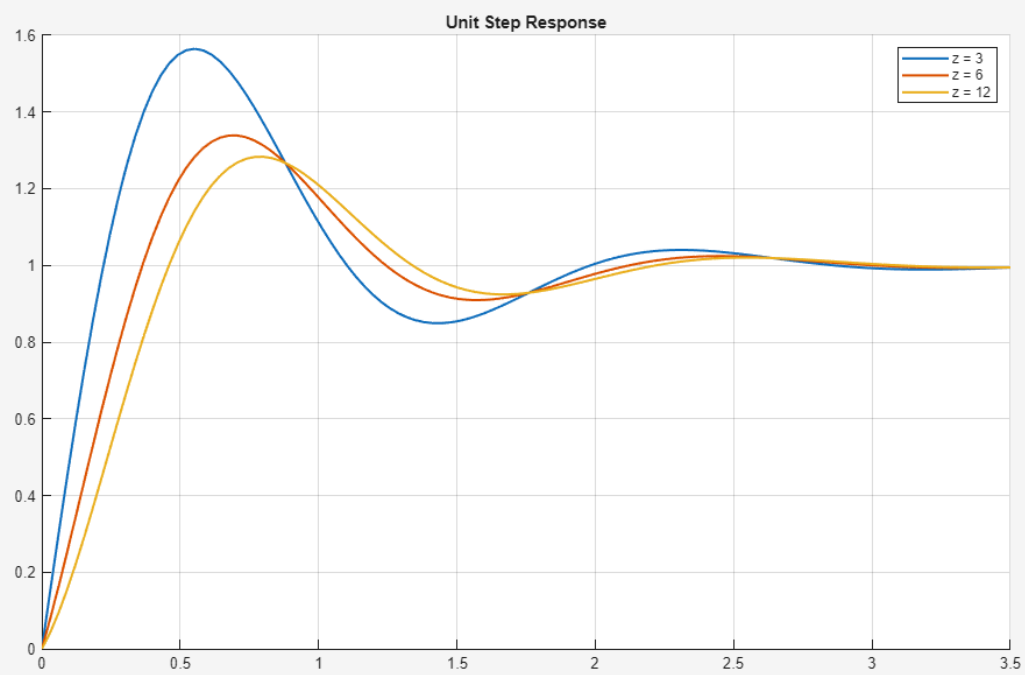
$$1 \times \left( \frac{1}{s^2 + 0.2s + 1} \right) = \frac{1}{s^2 + 0.2s + 1}$$

obtaining unit step response for transfer function 2

$$\frac{1}{s} \times \left( \frac{s}{s^2 + 0.2s + 1} \right) = \frac{1}{s^2 + 0.2s + 1}$$

3.

```
1 %% Elisha John D. Aton Problem #3
2 tf_denominator = [1 3 15];
3 z_values = [3 6 12];
4
5 % Unit step
6 figure; hold on; grid on;
7 for z = z_values
8     tf_numerator = (15/z)*[1 z];
9     sys = tf(tf_numerator, tf_denominator);
10    [y, t] = step(sys);
11    plot(t, y, 'LineWidth', 1.5);
12 end
13 title('Unit Step Response');
14 legend('z = 3', 'z = 6', 'z = 12');
15
16 % Unit impulse
17 figure; hold on; grid on;
18 for z = z_values
19     tf_numerator = (15/z)*[1 z];
20     sys = tf(tf_numerator, tf_denominator);
21     [y, t] = impulse(sys);
22     plot(t, y, 'LineWidth', 1.5);
23 end
24 title('Unit Impulse Response');
25 legend('z = 3', 'z = 6', 'z = 12');
```



4.

$$y'' + 4y' + 4y = u$$

$$y(0) = 0$$

$$y'(0) = 0$$

by laplace transform

$$y'' + 4y' + 4y = u$$

$$Y(s^2 + 4s + 4) = \frac{1}{s}$$

$$Y = \frac{1}{s(s+2)^2}$$

by partial fractions

$$\left( \frac{1}{s(s+2)^2} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{(s+2)^2} \right) s(s+2)^2$$

$$1 = A(s+2)^2 + Bs(s+2) + Cs$$

$$Y = \frac{1/4}{s} - \frac{1/4}{s+2} - \frac{1/2}{(s+2)^2}$$

by inverse laplace

$$y(t) = \frac{1}{4} - \frac{1}{4}e^{-2t} - \frac{1}{2}te^{-2t}$$

Let  $s = 0$

$$1 = A \cdot 4 = A = \frac{1}{4}$$

Let  $s = -2$

$$1 = -2C \Rightarrow C = -\frac{1}{2}$$

Let  $s = 1$

$$1 = 9A + 3B + C \Rightarrow 3B = 1 - \frac{9}{4} + \frac{1}{2}$$

$$B = -\frac{1}{4}$$

```

1 %% Elisha John D. Aton Problem #4
2 eqn = tf(1, [1 4 4]);
3
4 % Step response from MATLAB
5 [y_eqn, t] = step(eqn);
6 % Analytical solution
7 solved_eqn = 1/4 - (1/4)*exp(-2*t) - (1/2)*t.*exp(-2*t);
8
9 % Plotting
10 figure; hold on; grid on;
11 plot(t, y_eqn, 'b-', 'LineWidth', 2);
12 plot(t, solved_eqn, 'r--', 'LineWidth', 2);
13 title('Step Response: Analytical vs step()');
14 xlabel('Time (s)'); ylabel('y(t)');
15 legend('step()', 'Analytical');
```

